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SUPERSEDING
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MILITARY SPECIFICATION

FILM, OSCILLOGRAPH/INSTRUMENT RECORDING

BLACK AND WHITE

This specification is approved for use within the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers film used for recording high resolution imagery from Cathode ray tubes and optical instrumentation emitting the P-11 phosphor spectrum. The film will be capable of rapid processing to a negative image.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Boxes, Folding, Paperboard
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	-	Boxes, Shipping, Fiberboard
PPP-B-640	-	Boxes, Fiberboard, Corrugated, Triple-Wall
PPP-B-676	-	Boxes, Setup

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Ogden ALC/MMEDO, Hill AFB UT 84056 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	-	Preservation, Methods of
MIL-D-4824	-	Developers, Photograhic, Armed Forces

STANDARDS

FEDERAL

FED-STD-170	-	Film, Photographic, Black and White Classification and Testing Methods

MILITARY

MS 26565	-	Spool-Photograhic Film, Aerial
MS 27575	-	Container, Plastic, Photographic Aerial Film, Outline of

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 882	-	Tensile Properties of Thin Plastic Sheeting

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

PH1.2	-	Dimensions of Aerial Film Spools
PH1.10	-	Dimensions for Unperforated and Perforated Photographic Film in Rolls, Including Leaders and Trailers, for Aerial and Related Users
PH1.25	-	Specifications for Safety Photographic Film
PH1.31	-	Method for Determining the Brittleness of Photographic Film
PH1.32	-	Methods for Determining the Dimensional Change Characteristics of Photographic Films and Papers

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PH1.33	-	Dimensions for 16mm 100-Foot, 16mm 200-Foot, 35mm 100-Foot, and 70mm 100-Foot Spools for Recording Instruments and for Microfilm and Still-Picture Cameras
PH1.46	-	Dimensions of Containers for Packaging and Storage of Aerial Photographic Film on Spools
PH2.19	-	Conditions for Diffuse and Doubly Diffuse Transmission Measurements (Transmission Density)
PH2.29	-	Simulated Daylight Source for Photographic Sensitometry
PH2.33	-	Method for Determining the Resolving Power of Photographic Materials
PH2.34	-	Method for Determining the Speeds of Monochrome Photographic Negative Films for Aerial Photography
PH22.110	-	Dimensions for 16-mm Motion-Picture Film Perforated 2R

(Application for copies should be addressed to the American Standards Institute, 1430 Broadway, New York, NY 10018.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail requirements. Numerical requirements and allowable deviations shall be as specified in the appropriate table.

3.1.1 Film edge marking. The film shall be sequentially numbered along one edge of the film within .075 inch [1.905 millimeters (mm)] of the film edge and at intervals specified in Section F of the appropriate table. Manufacturer's commercial marking will be permitted provided such marking falls within the specified limit and does not interfere with the edge numbering.

3.2 Sensor compatibility. The film supplied under the specification shall not tear or be attributed as the cause of a camera sensor malfunction or failure, when the film is tested as outlined in 4.2.1.1.

3.3 Machine processor compatibility. The film shall show no evidence of defects attributable to manufacturing, spooling or packaging which could affect the extraction of intelligence information when tested as specified in 4.2.1.2.

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3.4 Physical characteristics. The film shall meet the requirements specified under Section A of the appropriate table.

3.5 Sensitometric characteristics. The film shall meet the requirements specified in Section B of the appropriate table when tested as specified in 4.2.2.

3.5.1 Density. Density for the purpose of this specification shall be defined as standard diffused visual density, as described in ANSI PH 2.19.

3.6 Dimensional stability. The film shall not vary dimensionally by an amount greater than the indicated maximum permissible change under Section C of the appropriate table.

3.7 Spools and containers. Spools and containers provided with the film shall meet the requirements of Section D and E of the appropriate table.

3.8 Film quality. The film quality shall be in accordance with high grade commerical practice covering the manufacturer of sensitized photographic materials. The manufacturer's commercial warranty shall apply. The manufacturer's liability shall be limited to replacement of the defective material or refund of the purchase cost, at the discretion of the Government.

3.9 Bid sample. Unless otherwise specified in the contract or purchase order, a bid sample of five rolls of film must be submitted for quality assurance testing as specified in 4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the quality control procedures which insure performance of the product in accordance with the requirements specified in Section 3 of this document. The Government reserves the right to perform any of the tests set forth in the specification where such testing is deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Test methods.

4.2.1 Compatibility tests. The compatibility tests specified herein shall be run by the procuring activity. The test results shall be provided to contractors to satisfy DCAS requirements.

4.2.1.1 Sensor test. A minimum of two rolls of film shall be run in two operational sensors specified in Section G of the appropriate table to assure that the system will transport the

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film and that the film does not cause camera/sensor malfunction. The units shall be run at ambient temperatures specified in Section G of the appropriate table and shall be cycled through high and low cycling rates. Failure to meet the requirements as specified in 3.2 shall be cause for rejection.

4.2.1.2 Machine processing test. Samples of the film shall be processed in the processing machine specified in Section H at the temperatures and transport rates specified in Section H of the appropriate table. Unless otherwise specified, high contrast chemistry appropriate for the film shall be used for this test. The dryer shall be 142° Fahrenheit (F) [61.1 degrees Celsius (°C)] maximum with ambient air temperature of 70° ± 2° F (21.11 ± 1.11°C) and 50% ± 5% relative humidity. The processed film shall be evaluated on a light table by viewing with not more than 10X magnification. Failure to meet the requirements as specified in 3.3 shall be cause for rejection.

4.2.2 Sensitometric test. Failure of the results of the sensitometric tests to meet the requirements specified in 3.5 shall be cause for rejection.

4.2.2.1 Exposure. To determine the sensitometric characteristics, the film exposure shall be between 1/100 and 1/300 second with exposure modulation as described in Federal Standard 170. The light source shall conform to the requirements of ANSI PH 2.29 except that a Wratten 47 filter shall be positioned in the light path. Computations of the absolute log exposure resulting from the use of this method, shall take into consideration the light attenuated by the Wratten 47 filter.

4.2.2.2 Processing. To determine the sensitometric characteristics, the film must be developed in Armed Forces Developer #1 of MIL-D-4824 for eight minutes at 68° F (20°C) and fixed in accordance with FED-STD-170.

4.2.2.3 Effective recording film speed. The effective recording film speed shall be defined as the value determined by the methods of all paragraphs of ANSI PH 2.34 except 3.3.2.4 "Light Source", and 3.3.3 "Processing".

4.2.3 Physical characteristics test. Physical characteristics shall be tested in accordance with the methods in Section A of the appropriate table. Failure to meet the requirements as specified in 3.4 shall be cause for rejection.

4.2.4 Dimensional stability test. Dimensional stability shall be tested in accordance with the methods specified in Section C of the appropriate table. Failure to meet the requirements as

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specified in 3.6 shall be cause for rejection.

TABLE I

(FILM FOR RECORDING HIGH RESOLUTION IMAGERY FROM THE CATHODE RAY TUBE, EMITTING THE P-11 PHOSPHOR SPECTRUM, OF THE AN/AAS-18 INFRARED DETECTING SET.)

CHARACTERISTICS	REQUIREMENTS	TEST METHOD OR STANDARD
A. Physical Characteristics		
Thickness	4.5 mils ± .5 mils [114.3±12.7 micrometers (μm)]	Micrometer Measurement
Width	4.96 in ± .01 [12.598±0.025 centimeters (cm)]	ANSI PH 1.10
Length	350 ft minimum [106.68 meters (m)]	Measurement
Cut of Lead End		ANSI PH 1.10
Film Base		ANSI PH 1.25
Tensile Strength	Minimum 9,500 lbs [133.95 kilograms (kg)]	ASTM D882
Brittleness	Maximum Value .445 inch (11.303 mm)	Method B, ANSI PH 1.31
B. Sensitometric Characteristics		
Light Source	Daylight with Wratten 47	ANSI PH 2.29
Effective Recording Film Speed	750 to 1500	4.2.2.3
Gamma	1.2 to 2.4	FED-STD-170
Maximum Density	2.4 Minimum	ANSI PH 2.19
Base Plus Fog Density	.20 Maximum	FED-STD-170
Resolution		
Total Object Contrast		
1000:1	100 lines/mm minimum	ANSI PH 2.33
1.6:1	42 lines/mm minimum	ANSI PH 2.33
C. Dimensional Stability (Maximum Permissible Change)		
Humidity Coefficient of Expansion % of % RH	.003%	ANSI PH 1.32
Thermal Coefficient of Linear Expansion	.0015%	ANSI PH 1.32
Dim change Due to Processing	.05%	ANSI PH 1.32
Dim Change Due to Processing & Aging	.1%	ANSI PH 1.32

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D. Spool Number ANSI or MS	9 -9	ANSI PH 1.2 MS 26565
E. Container Number ANSI or MS	11 -9 -9	ANSI PH 1.46 MS27575
F. Film Edge Marking (Sequential Numbering)	Intervals not greater than 12 inches (0.3048 m) 3.1.1	
G. Sensor Compatibility		
AN/AAS-18 Camera System Ambient Temperature	3.2 100°F (37.77°C)	4.2.1.1
H. Machine Processing Compatibility		
Roller Transport Type EH-38A Temperature Transport Rate	3.3 90°±1°F (32.22±0.55°C) 7±2 feet per minute (2.1336±0.6096 m/min)	4.2.1.2

TABLE II

(FILM FOR RECORDING HIGH RESOLUTION IMAGERY FROM THE CATHODE RAY TUBE, EMITTING THE P-11 PHOSPHOR SPECTRUM, OF A HIGH SPEED CAMERA).

CHARACTERISTICS	REQUIREMENT	TEST METHOD OR STANDARD
A. Physical Characteristics		
Thickness	4.5 mils nominal (114.3 µm)	Micrometer Measurement
Width	15.95±0.03mm	ANSI PH 22.110
Length	125 ft (38.1 m)minimum	
Perforations	Double perf (high speed)	ANSI PH 22.110
Film Base	Polyester	
Tensile Strength	Minimum 9,500 lbs (133.95 kg)	ANSI PH 1.25 ASTM D882
Brittleness	Maximum Value .445 inch (11.303 mm)	Method B, ANSI PH 1.31

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B. Sensitometric Characteristics

Light Source	Daylight plus W47 Filter	ANSI PH 2.29
Effective Recording Film Speed	750 to 1500	4.2.2.3
Gamma	1.2 to 2.4	FED-STD-170
Max Density	2.4 Minimum	ANSI PH 2.19
Base plus Fog Density	.20 maximum	FED-STD-170
Resolution		
Total Object Contrast		
1000:1	100 lines/mm minimum	ANSI PH 2.33
1.6:1	42 lines/mm minimum	ANSI PH 2.33

C. Dimensional Stability (Maximum Permissible Change)

Humidity Coefficient of Expansion % of % RH	.003%	ANSI PH 1.32
Thermal Coefficient of Linear Expansion	.0015%	ANSI PH 1.32
Dim Change Due to Processing	.05%	ANSI PH 1.32
Dim Change Due to Processing and Aging	.1%	ANSI PH 1.32

D. Spools — ANSI PH 1.33

E. Containers — In accordance with standard commerical practice.

F. Film Edge Marking

(Sequential Numbering) — Intervals not greater than 12 in. (0.3048 m) 3.1.1

G. Sensor Compatibility

Maverick TGM Recorder	3.2	4.2.1.1
Ambient Temperature	100°F (37.77°C)	

H. Machine Processing Compatibility

Prostar Film Processor	3.3	4.2.1.2
Prostar Chemistry		

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Temperature	90°±1°F (32.22±0.55°C)	
Transport Rate	5 feet per minute (1.524 m/min)	
A N D		
Roller Transport		
Type EH-38A	3.3	4.2.1.2
Temperature	90°±1°F (32.22±0.55°C)	
Transport Rate	7 feet per minute (2.1336 m/min)	

5. PACKAGING

5.1 Preservation and Packaging. Film shall be preserved and packaged Level A or C as specified.

5.1.1 Level A. Film on spools and in containers as provided in Sections D and E of the appropriate table shall be preserved and packaged Method IA-8 in accordance with MIL-P-116.

5.1.2 Level C. Film on spools and in containers as provided in Sections D and E of the appropriate table shall be packaged to insure protection against damage from supply source to the first receiving activity.

5.2 Packing. Film as packaged in 5.1.1 and 5.1.2 shall be packed level A,B, or C as specified.

5.2.1 Level A. Film packaged as specified in 5.1.1 will be packed in containers conforming to PPP-B-566, PPP-B-676 or PPP-B-636 and will serve as the intermediate container. Six or more intermediate containers will be packed into containers conforming to PPP-B-601, Type Overseas, Style A or PPP-B-621, Class 2, Style 2 or 4. Closure will be in accordance with the appendix of the particular box specification.

5.2.2 Level B. Film packaged as specified in 5.1.1 will be packed in intermediate containers conforming to PPP-B-566, PPP-B-676 or PPP-B-636. Six or more intermediate containers will be packed in containers conforming to PPP-B-636, Class weather-resistant, Variety DW (doublewall) or PPP-B-640. Closure will be in accordance with the appendix to the particular box specification.

5.2.3 Level C. Film packed as specified in 5.1.2 shall be packed to insure protection against damage from transportation from supply source to the first receiving activity. It shall also meet the rules and regulations of the Department of Transportation, as applicable to the mode(s) of transportation to be utilized.

6. NOTES

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6.1 Intended use. The film covered by this specification is intended for use in recording high resolution imagery from optical instrumentation.

6.2 Ordering data. Procurement document should specify:

a. Title, number, and date of this specification.

b. Quantity required.

c. Applicable levels of preservation, packaging, and packing required (see 5.1, 5.2).

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Air Force-99

Preparing Activity:
Air Force-70

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